Statistics & Probability

ANSWER KEY

Section 1: Permutations & Combinations

1 In how many ways can a president and a vice president be chosen from a committee of 6 people?

a. 30 ✓

b. 15

c. 36

d. 12

Order matters (roles are distinct): $P(6, 2) = 6 \times 5 = 30$.

2 How many distinct committees of 3 students can be selected from a group of 7 students?

a. 35 ✓

b. 210

c. 21

d. 42

Order does not matter: $\binom{7}{3} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$.

3 What is the value of $\frac{6!}{4! \times 2!}$?

a. 15 ✓

b. 30

c. 12

d. 20

$\frac{6 \times 5 \times 4!}{4! \times 2} = \frac{30}{2} = 15$.

Section 2: Conditional & Compound Probability

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4 If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, what is $P(A \cup B)$?

a. 0.8 ✓

b. 0.7

c. 0.9

d. 1.1

Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.3 = 0.8$.

5 If $P(B) = 0.4$ and $P(A \cap B) = 0.12$, what is the conditional probability $P(A|B)$?

a. 0.30 ✓

b. 0.48

c. 0.25

d. 0.52

$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.12}{0.40} = 0.30$.

6 If events A and B are independent with $P(A) = 0.3$ and $P(B) = 0.4$, what is $P(A \cap B)$?

a. 0.12 ✓

b. 0.70

c. 0.10

d. 0.24

For independent events: $P(A \cap B) = P(A) \times P(B) = 0.3 \times 0.4 = 0.12$.

Section 3: Normal Distribution & Empirical Rule

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7 A normal distribution has mean $\mu = 50$ and standard deviation $\sigma = 5$. What percentage of data falls between 45 and 55?

a. 68% ✓

b. 95%

c. 99.7%

d. 50%

The interval $[45, 55]$ represents $[\mu - \sigma, \mu + \sigma]$. By the empirical rule, exactly 68% of the data lies within 1 standard deviation.

8 In a normal distribution with mean 100 and standard deviation 10, what is the probability that a value is greater than 120?

a. 2.5% ✓

b. 5%

c. 16%

d. 0.15%

$120 = \mu + 2\sigma$. The middle 95% lies within 2σ. The remaining 5% is split into two tails: 2.5% in each tail.

9 What is the z-score for a value $x = 85$ from a distribution with mean $\mu = 70$ and standard deviation $\sigma = 10$?

a. 1.5 ✓

b. 1.2

c. 2.0

d. 0.85

$z = \frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$.

Section 4: Measures of Dispersion & Variance

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10 If the variance of a data set is 36, what is its standard deviation?

a. 6 ✓

b. 18

c. 72

d. 12

Standard deviation $\sigma = \sqrt{\text{Variance}} = \sqrt{36} = 6$.

11 If 5 is added to every value in a dataset, how does the standard deviation change?

a. **It remains unchanged** ✓

b. It increases by 5

c. It increases by 25

d. It multiplies by 5

Adding a constant shifts all data points equally, leaving the spread (standard deviation and variance) unchanged.

12 What is the range of the data set: $12, 5, 23, 18, 9, 31, 14$?

a. 26 ✓

b. 25

c. 24

d. 31

Range $= \text{Maximum} - \text{Minimum} = 31 - 5 = 26$.

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